Customer Churn Analysis and Prediction Using Machine Learning

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1. Introduction

Customer churn refers to customers who stop using a company's service. This project aims to predict whether a customer will churn or not using supervised machine learning algorithms.

2. Technologies and Libraries Used

- Python
- Pandas – Data manipulation
- NumPy – Numerical operations
- Matplotlib & Seaborn – Data visualization
- Scikit-learn – Machine learning algorithms
- Train-test split for data splitting
- StandardScaler for feature scaling
- LabelEncoder for categorical encoding
- GridSearchCV for hyperparameter tuning

3. Project Workflow

- Data Collection and Loading
- Data Cleaning and Handling Missing Values
- Exploratory Data Analysis (EDA)
- Encoding Categorical Variables
- Feature Scaling
- Train-Test Split
- Model Training (Logistic Regression, KNN, Decision Tree, Random Forest, SVM)
- Model Evaluation using Accuracy, Confusion Matrix, Classification Report
- Hyperparameter Tuning using GridSearchCV
- Final Model Selection

4. Business Impact

This model helps companies identify customers who are likely to churn. Businesses can take preventive measures such as discounts or special offers to retain high-risk customers.

5. Viva Questions and Answers

- Q1: What is customer churn?
- A: Customer churn refers to customers who stop using a company’s service.
- Q2: Is this a regression or classification problem?
- A: It is a supervised classification problem.
- Q3: Why do we use StandardScaler?
- A: To normalize feature values so that models like KNN and SVM perform better.
- Q4: What is overfitting?
- A: Overfitting occurs when a model performs well on training data but poorly on unseen test data.
- Q5: Why use Random Forest?
- A: Random Forest reduces overfitting by combining multiple decision trees.
- Q6: What is GridSearchCV?
- A: It is a technique used to find the best hyperparameters for a model.
- Q7: What evaluation metrics did you use?
- A: Accuracy, Confusion Matrix, and Classification Report.
- Q8: Why is churn prediction important?
- A: Because retaining customers is more cost-effective than acquiring new ones.